

The Jones family has 4 children and want one more child to complete their family. $\frac{3}{4}$ of the children are boys and the rest are girls.

1. What is the probability that the next child will be a boy, represented as a percent rounded to the nearest hundredth?

$$\frac{1}{2} = 0.5 = 50\%$$

2. What is the theoretical probability that the next child will be a boy?

$$\frac{1}{2}$$

Describe how each of the following can be used to simulate if the next child will be a boy.

3. A spinner with 8 equal sections

Answers may vary.

Mark half the sections for "boy" and half for "girl"

4. A twelve-sided die

Answers may vary.

Even numbers represent boy, odd numbers represent a girl.

5. A bag with 60 marbles

Answers may vary.

Choose $\frac{1}{2}$ to represent a boy and $\frac{1}{2}$ to represent a girl.

The weather forecast says there is a 50% chance it will rain today.

Describe how each of the following can be used to simulate if it will rain today.

6. A coin

Heads – rain Tails – rain
 or
Tails – no rain Heads – no rain

7. Different numbers in a box

Answers may vary.

Even numbers – rain

Odd numbers – no rain



There has to be an even number of each.

There are 4 types of blood, A, B, AB, and O. Blood type is determined by genes inherited from parents. Suppose that a type AB blood donor is needed at a local hospital for a patient. 0.4 donors have type AB blood.

8. Find the probability that the next donor will not have type AB blood.

0.6

9. What is the theoretical probability the next donor will have type AB blood?

$\frac{1}{4}$

10. Describe a way to simulate the probability the next donor will have type AB blood.

Answers may vary.

Spinning a spinner with 4 equal sized sections, one for each blood type.